



Flowserve US Raleigh NC (US) Flowserve Cooling System Performance Report May 2026

31 May 2026

Location

Flowserve US Raleigh NC (US)

Prepared by

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1. System Health Check

Corrosion Control:

During the reporting period, the system remained under good control. The mild steel and copper corrosion rates averaged 1.06 mpy and 0.058 mpy, respectively, both of which are well within the recommended limits of <3 mpy and <0.5 mpy.

Scale Control:

The tagged polymer concentration was consistently maintained within the acceptable range, contributing to effective scale control within the system.

Microbial Control:

The ORP spikes were not consistent though there was biocide dosage into the system, which may be attributed to a potential measurement issue with the probe. To further investigate, probe inspection, Free Residual Chlorine (FRC) level testing, and a dip slide test will be carried out during the upcoming service visit.

2. Water Efficiency

During May, the tower water conductivity averaged 930 $\mu\text{S}/\text{cm}$ (5.1 cycles), remaining close to the target setpoint of $950 \pm 20 \mu\text{S}/\text{cm}$ (5.2 cycles). This indicates stable conductivity control and effective cycles management, supporting improved water conservation within the system.

3. Product Efficiency

The product concentration was well maintained within defined threshold limit for entire reporting period indicating generally stable product control. Product consumption was 24.6 lbs. in May, which matched with actual consumption.

4. Proactive System Support

Ackumen records an average of six alarms per controller overall, whereas this site generated only two alarms during the reporting period. These alarms were associated with the oxidant dosing pump and were primarily attributed to pump-related issues. An increase in copper corrosion rate was also observed, likely due to excessive oxidant dosing. These deviations were linked to temporary operational and chemical feed variations and were effectively resolved during the latter part of the reporting period.

The overall Goodness of Control remained at 98%, based on the defined threshold control ranges for key cooling system performance parameters.

Performance Summary

Goodness of Control

Parameter	Context Point	Last Reading	Average	Standard Deviation	Minimum Value	Maximum Value	Lower Limit	Upper Limit	Count Below	Count Above	% In Range	Gauge Chart
Corrosion Probe - MS(MPY)	ACM skid	1.12	1.06	0.133	0.776	2.67	0.00	3.00	0.00	0.00	100	
Corrosion Probe - Cu(MPY)	ACM skid	0.0114	0.457	1.50	0.00	9.45	0.00	0.500	0.00	307	90.0	
Traced Product(ppm)	ACM skid	85.0	95.1	27.0	76.5	283	60.0	130	0.00	150	95.1	
pH(pH)	ACM skid	8.42	8.40	0.135	7.98	8.63	7.50	9.50	0.00	0.00	100	
ORP(mV)	ACM skid	626	631	25.4	488	706	100	700	0.00	11.0	99.6	
Electrode Conductivity(μS/cm)	ACM skid	869	886	43.0	583	991	800	1,500	110	0.00	96.4	
Fluorometer - Turbidity(NTU)	ACM skid	1.64	2.65	1.25	0.226	27.1	0.00	75.0	0.00	0.00	100	

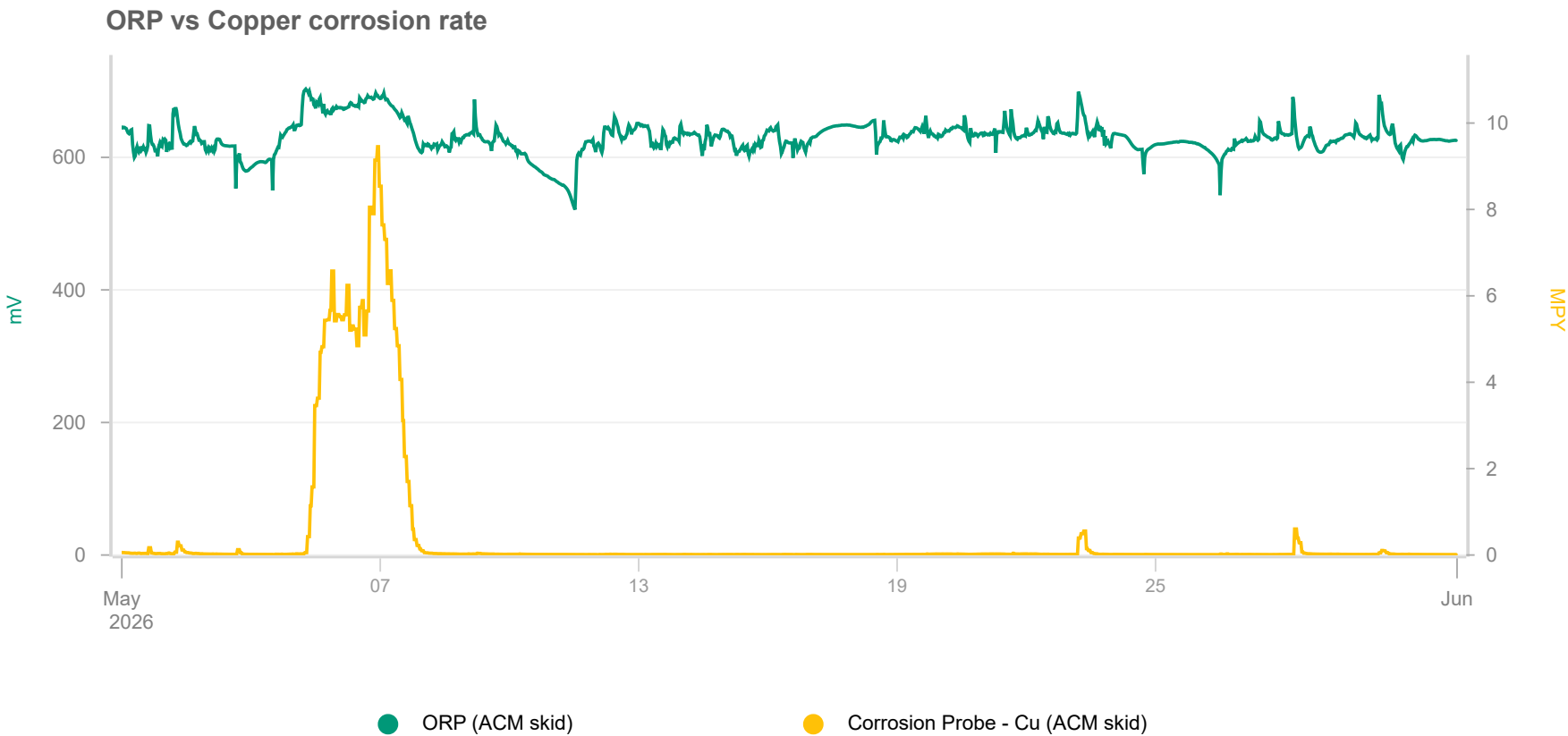
Comments

The control limits for each cooling water parameter in the performance summary have been defined in accordance with the proposed treatment program at the recommended cycles of concentration. Overall, the key performance indicators were well maintained throughout the reporting period.

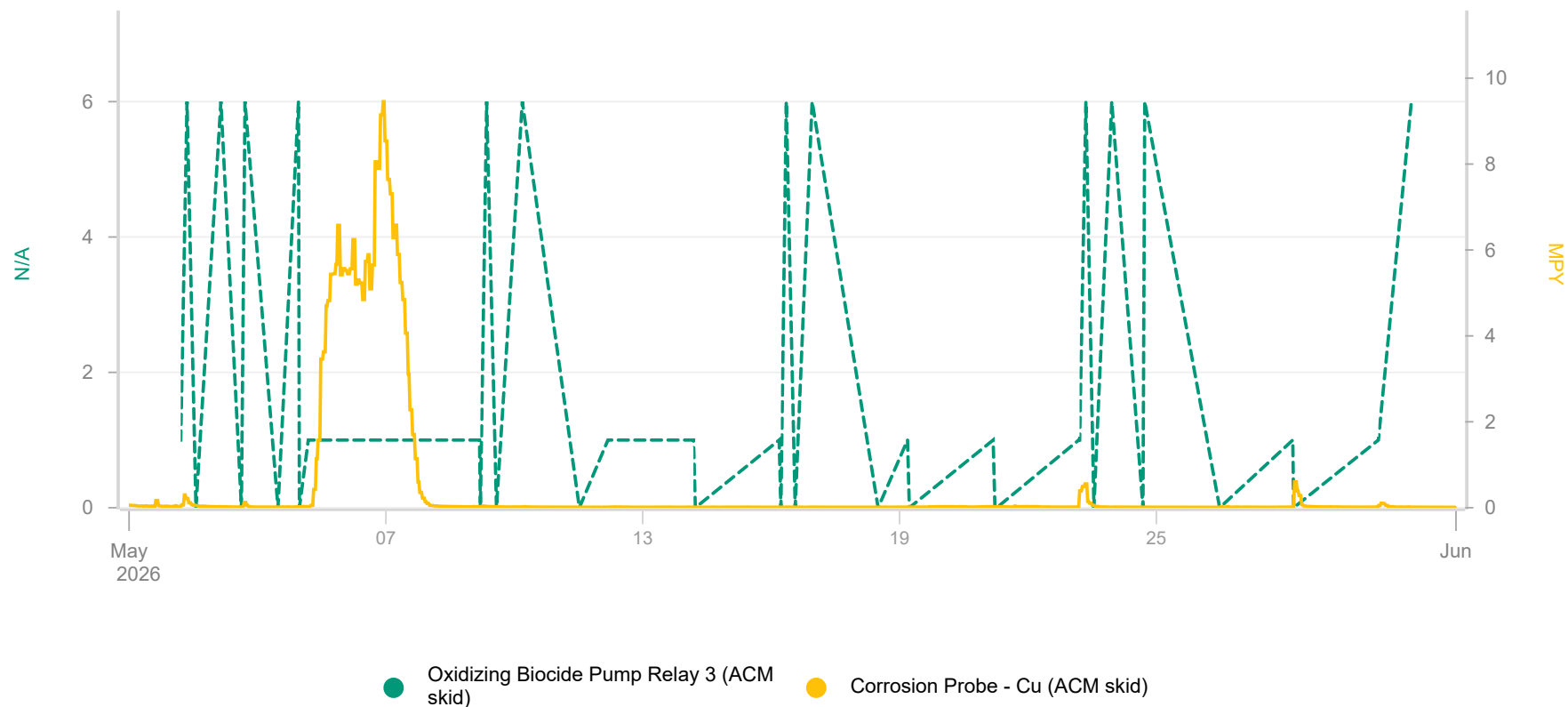
The copper corrosion rate averaged 0.45 mpy, approaching the acceptable threshold of 0.5 mpy. This increase was primarily attributed to elevated corrosion levels observed between 5th May and 8th May, likely due to operational factors such as excessive oxidant dosing.

Following this period, the system was stabilized, and corrosion levels were effectively brought under control for the remainder of the reporting period.

KPM trends



Oxidizing biocide pump relay status vs Copper corrosion rate



Comments

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